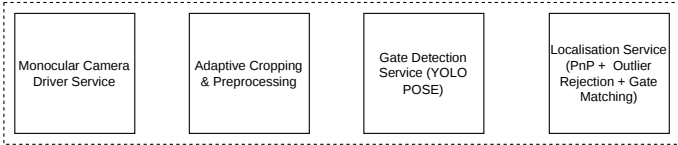
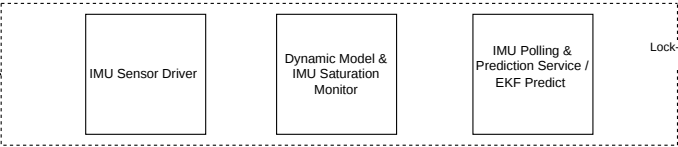


T1 : The Vision Pipeline (Runs at 30–90 Hz on Jetson GPU/CPU)

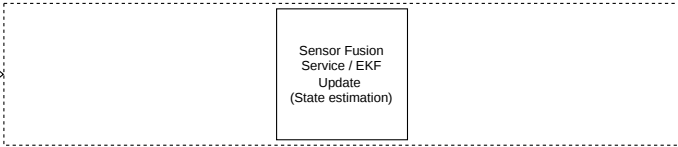


T2 : High-Frequency IMU & Prediction (Runs at 1,000–2,000 Hz on Jetson CPU)

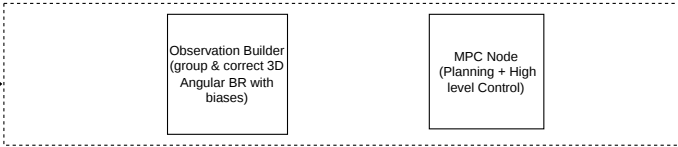


Lock-Free Queue (Timestamped Poses)
+ Delay Compensation

T3 : The Event-Driven EKF Update (Runs at 30–90 Hz on Jetson CPU)



T4 : The Planning & Control Loop (Runs at 50–100 Hz on Jetson CPU)



Shared Memory
with Mutex Locks :
16D State Vector (global)

1KHz UART: 4D Motor Speeds
+ 3D Linear Accel
+ RPM-Filtered 3D Body Rates

STM32 FLIGHT CONTROLLER (BETAFLIGHT)

50-100Hz UART :
CTBR Commands

T5 : Low-Level Hardware Loop (Runs at 1,000+ Hz on the Flight Controller)

